Conscious Agents and Markov Blankets: A Formal Correspondence

Sloan Austermann
OmniSciences

Abstract

We establish a formal correspondence between Hoffman's Conscious Agents and Friston's Markov Blanket systems. We construct a mapping $\Theta : \mathbf{Con} \to \mathbf{MB}$ and prove that the perception-decision-action kernel factorization characteristic of conscious agents implies the conditional independence property that defines Markov blankets (Theorem 3.1). We prove compositional correspondence for non-interacting agents (Theorem 4.3) and provide a complete characterization for interacting agents: the Blanket-Mediated Interaction Condition (BMIC) is necessary and sufficient for maintained dissociation (Theorems 4.6–4.8). We characterize the inverse mapping, showing precisely when a Markov blanket system admits conscious agent structure (Theorem 5.1). We connect integration in this framework to Tononi's Φ, proving that dissociability corresponds to zero integrated information (Theorem 6.3). Crucially, we show this correspondence captures genuine structure: the conditional independence property is non-generic, forming a measure-zero set with codimension at least $m^2b(e-1)$ in the space of stochastic matrices.

1 Introduction

Hoffman's Conscious Agent framework [6, 5] and Friston's Free Energy Principle with Markov blankets [4] represent two influential formal approaches to consciousness and cognition. Despite their different motivations—Hoffman approaching from perception theory, Friston from statistical physics—both employ similar mathematical machinery: state spaces, probabilistic transitions, and notions of boundaries between agent and environment. The structural affinity between these frameworks has been explored at the network level by Fields et al. [1], who developed a category-theoretic treatment of composed conscious agents and their communication channels.

This paper establishes that the similarities run deeper than surface analogy. We prove that the kernel factorization structure of conscious agents *implies* the conditional independence that defines Markov blankets. This is a derived result: Hoffman's framework was not designed with Markov properties in mind, yet the perception-decision-action architecture guarantees them. Where Fields et al. [1] analyzed the network structure of interacting agents, the present work focuses on the individual agent-to-blanket correspondence and provides formal proofs of the structural relationship.

Main contributions:

1. A well-defined mapping $\Theta : \mathbf{Con} \to \mathbf{MB}$ (Definition 2.8)
2. Proof that Θ preserves conditional independence (Theorem 3.1)
3. Compositional correspondence for non-interacting agents (Theorem 4.3)
4. Complete BMIC characterization: necessary and sufficient conditions for maintained dissociation under interaction (Theorems 4.6–4.8)
5. Characterization of the inverse mapping $\mathbf{MB} \to \mathbf{Con}$ (Theorem 5.1)
6. Connection to integrated information (Theorem 6.3)
7. Non-genericity analysis with precise codimension bounds (Section 7)

2 Formal Definitions

2.1 Measure-Theoretic Foundations

All state spaces are assumed to be standard Borel (measurably isomorphic to a Borel subset of a Polish space). This ensures regular conditional probabilities exist and are unique almost everywhere.

Definition 2.1 (Markov Kernel). *A* Markov kernel *from* (S_1, Σ_1) *to* (S_2, Σ_2) *is a function* $K : S_1 \times \Sigma_2 \to [0, 1]$ *such that:*

1. *For each* $s_1 \in S_1$, $K(s_1, \cdot)$ *is a probability measure on* S_2
2. *For each* $A \in \Sigma_2$, $K(\cdot, A)$ *is measurable*

We write $K[s_1, s_2]$ *for the density when it exists, and* $\mathcal{K}(S_1, S_2)$ *for the set of all Markov kernels from* S_1 *to* S_2.

Remark (Notation Convention). *Throughout this paper,* $K(s_1, A)$ *denotes the measure-theoretic form (probability that* K *maps* s_1 *into measurable set* A*), while* $K[s_1, s_2]$ *denotes the density with respect to a fixed reference measure. Brackets always denote densities.*

Definition 2.2 (Kernel Composition). *For* $K_1 \in \mathcal{K}(S_1, S_2)$ *and* $K_2 \in \mathcal{K}(S_2, S_3)$*:*

$$(K_2 \circ K_1)(s_1, A) = \int_{S_2} K_2(s_2, A)\, K_1(s_1, ds_2)$$

This is the Chapman-Kolmogorov equation.

Definition 2.3 (Kernel Tensor Product). *For* $K_1 \in \mathcal{K}(S_1, T_1)$ *and* $K_2 \in \mathcal{K}(S_2, T_2)$*:*

$$(K_1 \otimes K_2)[(s_1, s_2), (t_1, t_2)] = K_1[s_1, t_1] \cdot K_2[s_2, t_2]$$

2.2 Conscious Agents

Definition 2.4 (Conscious Agent). *A* conscious agent *is a tuple* $\alpha = (X, G, W, P, D, A)$ *where:*

State Spaces *(all standard Borel):*

- X*: Experience space (internal states)*
- G*: Action space (behavioral outputs)*
- W*: World state space (external environment)*

Markov Kernels:

- $P \in \mathcal{K}(W \times X, X)$*: Perception kernel*
- $D \in \mathcal{K}(X, G)$*: Decision kernel*
- $A \in \mathcal{K}(G \times W, W)$*: Action kernel*

The dynamics form a cycle $W \to X \to G \to W$ *with combined kernel:*

$$K_\alpha[(x, g, w) \to (x', g', w')] = P[w, x, x'] \cdot D[x', g'] \cdot A[g', w, w'] \tag{1}$$

Remark. *The kernel factorization encodes a specific causal ordering: world and experience determine next experience (via* P*), which determines action (via* D*), which with world determines next world state (via* A*). This perception-decision-action cycle is the defining characteristic distinguishing conscious agents from arbitrary dynamical systems.*

Example 2.5 (Simple Binary Agent). *Let* $X = G = W = \{0, 1\}$*. Define:*

$$P[w, x, x'] = \begin{cases} 0.9 & \text{if } x' = w \\ 0.1 & \text{if } x' \neq w \end{cases} \qquad \text{(noisy perception)}$$

$$D[x, g] = \begin{cases} 0.8 & \text{if } g = x \\ 0.2 & \text{if } g \neq x \end{cases} \qquad \text{(noisy identity)}$$

$$A[g, w, w'] = \begin{cases} 0.7 & \text{if } w' = g \\ 0.3 & \text{if } w' \neq g \end{cases} \qquad \text{(noisy action)}$$

This agent tends to perceive the world state, act according to its perception, and influence the world toward its action. The combined kernel K_α is an 8×8 stochastic matrix.

2.3 Markov Blanket Systems

Definition 2.6 (Markov Blanket System)**.** *A* Markov blanket system *is $M = (\Omega, \mu, s, a, \eta, K)$ where:*

State Space: *$\Omega = S_\mu \times S_s \times S_a \times S_\eta$ with partition:*

- *S_μ: Internal states (μ)*
- *S_s: Sensory states (s)*
- *S_a: Active states (a)*
- *S_η: External states (η)*
- *$B = (s, a)$: The blanket*

Dynamics: *$K \in \mathcal{K}(\Omega, \Omega)$ satisfying the* Markov blanket property*:*

$$P(\mu' \mid \mu, B, \eta) = P(\mu' \mid \mu, B) \tag{2}$$

The blanket B screens off internal from external: given μ and B, additional knowledge of η provides no information about μ'. This formalizes the intuition that the blanket mediates all interactions between interior and exterior. Fields [2] has argued that any physical interaction surface between spatially separated systems formally satisfies this property, a point we return to in Section 7.

2.4 The Mapping Θ

Definition 2.7 (Sensory Equivalence)**.** *For agent α and experience $x \in X$, world states $w_1, w_2 \in W$ are* sensorily equivalent given *x, written $w_1 \sim_x w_2$, if:*

$$P[(w_1, x), \cdot] = P[(w_2, x), \cdot] \quad \textit{as measures on } X$$

The sensory state function *$\sigma : W \times X \to S$ maps (w, x) to its equivalence class $[w]_{\sim_x}$.*

Remark. *Two world states are sensorily equivalent if they produce identical perception distributions. The agent cannot distinguish them phenomenologically—they are "experientially indistinguishable" from the agent's perspective.*

Definition 2.8 (The Mapping Θ)**.** *Given conscious agent $\alpha = (X, G, W, P, D, A)$, define:*

$$\Theta(\alpha) = (\Omega_\alpha, S_\mu, S_s, S_a, S_\eta, K_\alpha)$$

where:

- *$\Omega_\alpha = X \times G \times W$*
- *$S_\mu = X$ (internal = experience)*
- *$S_a = G$ (active = action)*
- *$S_\eta = W$ (external = world)*
- *$S_s = S$ (sensory = equivalence classes)*
- *K_α as in equation* (1)

For state $\omega = (x, g, w)$: $\mu = x$, $a = g$, $\eta = w$, $s = \sigma(w, x)$, $B = (\sigma(w, x), g)$.

Remark (Sensory State Dependence on Internal State)**.** *The sensory state $s = \sigma(w, x)$ depends on both the world state w and the internal state x. This differs from the standard Friston formulation, where sensory states depend only on external and blanket states. The dependence on x is philosophically appropriate: the agent's sensory window may depend on its current experiential state (e.g., attention modulates perception). Notably, this design choice finds support in the quantum extension of the Free Energy Principle [3], where the sensory state is determined*

by a quantum channel that depends on the system's reference frame—i.e., its internal state. The Markov blanket property still holds (Theorem 3.1), and σ functions as a sufficient statistic for the transition $w \mapsto x'$ given x: all information in w relevant to the next experience x' is captured by $\sigma(w, x)$.

Example 2.9 (Applying Θ to the Binary Agent)**.** *For the agent in Example 2.5, since $P[w, x, x']$ depends on w (specifically, on whether $x' = w$), we have $w_1 \not\sim_x w_2$ for $w_1 \neq w_2$. Thus $\sigma(w, x) = w$ and $S = W = \{0, 1\}$.*

The resulting MB system has:

- *$\Omega = \{0, 1\}^3$ (8 states)*
- *Internal $\mu = x$, sensory $s = w$, active $a = g$, external $\eta = w$*
- *Blanket $B = (w, g)$*

The Markov property $P(x' \mid x, w, g, w) = P(x' \mid x, w, g)$ holds trivially since $\eta = s = w$.

3 Main Theorem: Conditional Independence

Theorem 3.1 (Conditional Independence Correspondence)**.** *Let $\alpha = (X, G, W, P, D, A)$ be a conscious agent. Then $\Theta(\alpha)$ satisfies the Markov blanket property:*

$$P(\mu' \mid \mu, B, \eta) = P(\mu' \mid \mu, B)$$

where $\mu = x$, $\mu' = x'$, $B = (\sigma(w, x), g)$, and $\eta = w$.

Proof. We proceed in four steps.

Step 1: Derive $P(x' \mid x, g, w)$. From the dynamics kernel K_α, marginalize over g' and w':

$$P(x' \mid x, g, w) = \int_G \int_W P[w, x, x'] \cdot D[x', g'] \cdot A[g', w, w'] \, dg' \, dw'$$

Step 2: Apply normalization. Since $P[w, x, x']$ does not depend on g' or w', we may factor it out. (The interchange of integration order is justified by the Fubini-Tonelli theorem, since all kernel densities are non-negative measurable functions.)

$$P(x' \mid x, g, w) = P[w, x, x'] \cdot \int_G D[x', g'] \left(\int_W A[g', w, w'] \, dw' \right) dg'$$

The inner integral $\int_W A[g', w, w'] \, dw' = 1$ since $A[g', w, \cdot]$ is a probability measure on W. Similarly, $\int_G D[x', g'] \, dg' = 1$ since $D[x', \cdot]$ is a probability measure on G. Therefore:

$$P(x' \mid x, g, w) = P[w, x, x'] \cdot 1 \cdot 1 = P[w, x, x']$$

Step 3: Key observations.

- x' does not depend on the current action g
- x' depends on w only through the perception kernel $P[w, x, x']$

Step 4: Apply sensory equivalence. If $w_1 \sim_x w_2$, then $P[w_1, x, x'] = P[w_2, x, x']$ for all x'. Therefore $P(x' \mid x, g, w)$ depends on w only through $\sigma(w, x)$.

Since $s = \sigma(w, x)$ and x' is independent of g:

$$P(x' \mid x, s, g, w) = P(x' \mid x, s) = P(x' \mid x, B)$$

This is the Markov blanket property: $B = (s, a)$ screens off η from μ'. □

Corollary 3.2 (Boundary Correspondence)**.** *The blanket $B = (s, a)$ captures exactly the agent-world interface:*

- *Sensory side: P provides $w \to x'$ (input)*
- *Active side: A provides $g \to w'$ (output)*

4 Compositional Correspondence

The question of how conscious agents compose has been explored categorically by Fields et al. [1], who model agent networks via shared world-channels. Here we address a different but related question: under what conditions does the Θ mapping respect composition? We characterize the answer completely.

4.1 Non-Interacting Agents

Definition 4.1 (Non-Interacting)**.** *Agents α and β are* non-interacting *if $\alpha \otimes \beta$ has factored kernels:*

$$P_{\alpha\otimes\beta} = P_\alpha \otimes P_\beta, \quad D_{\alpha\otimes\beta} = D_\alpha \otimes D_\beta, \quad A_{\alpha\otimes\beta} = A_\alpha \otimes A_\beta$$

Lemma 4.2 (Tensor Products of Kernel Compositions)**.**

$$(P_1 \otimes P_2) \cdot (D_1 \otimes D_2) \cdot (A_1 \otimes A_2) = (P_1 \cdot D_1 \cdot A_1) \otimes (P_2 \cdot D_2 \cdot A_2)$$

Proof. Let $\omega_1 = (x_1, g_1, w_1)$, $\omega_2 = (x_2, g_2, w_2)$ and $\omega_1' = (x_1', g_1', w_1')$, $\omega_2' = (x_2', g_2', w_2')$. The left side evaluates to:

$$\begin{aligned}
&[(P_1 \otimes P_2) \cdot (D_1 \otimes D_2) \cdot (A_1 \otimes A_2)][(\omega_1, \omega_2) \to (\omega_1', \omega_2')] \\
&= (P_1 \otimes P_2)[(w_1, x_1, w_2, x_2), (x_1', x_2')] \cdot (D_1 \otimes D_2)[(x_1', x_2'), (g_1', g_2')] \\
&\quad \cdot (A_1 \otimes A_2)[(g_1', w_1, g_2', w_2), (w_1', w_2')] \\
&= P_1[w_1, x_1, x_1'] \cdot P_2[w_2, x_2, x_2'] \cdot D_1[x_1', g_1'] \cdot D_2[x_2', g_2'] \cdot A_1[g_1', w_1, w_1'] \cdot A_2[g_2', w_2, w_2']
\end{aligned}$$

Regrouping by agent:

$$\begin{aligned}
&= \big(P_1[w_1, x_1, x_1'] \cdot D_1[x_1', g_1'] \cdot A_1[g_1', w_1, w_1']\big) \cdot \big(P_2[w_2, x_2, x_2'] \cdot D_2[x_2', g_2'] \cdot A_2[g_2', w_2, w_2']\big) \\
&= K_{\alpha_1}[\omega_1 \to \omega_1'] \cdot K_{\alpha_2}[\omega_2 \to \omega_2'] = (K_{\alpha_1} \otimes K_{\alpha_2})[(\omega_1, \omega_2) \to (\omega_1', \omega_2')]
\end{aligned}$$

□

Theorem 4.3 (Compositional Correspondence: Non-Interacting)**.** *For non-interacting agents α and β:*

$$\Theta(\alpha \otimes \beta) \cong \Theta(\alpha) \otimes_{\mathbf{MB}} \Theta(\beta)$$

Proof. **Step 1:** State spaces: $\Omega_{\alpha\otimes\beta} \cong \Omega_\alpha \times \Omega_\beta$ via canonical reordering.

Step 2: Sensory factorization: By factorization of $P_{\alpha\otimes\beta}$, two world states (w_1, w_2) and (w_1', w_2') are sensorily equivalent given (x_1, x_2) iff both $w_1 \sim_{x_1} w_1'$ and $w_2 \sim_{x_2} w_2'$. Therefore:

$$s_{\alpha\otimes\beta}((w_1, w_2), (x_1, x_2)) = (s_\alpha(w_1, x_1), s_\beta(w_2, x_2))$$

Step 3: Kernel correspondence: $K_{\alpha\otimes\beta} = K_\alpha \otimes K_\beta$ by Lemma 4.2.

Step 4: Markov property: each factor satisfies it independently by Theorem 3.1. □

4.2 Interacting Agents and BMIC

When agents interact, their kernels may fail to factor. We classify interactions by which kernel is affected.

Definition 4.4 (Interaction Classification)**.** *For agents α and β, we classify interactions as:*

1. ***Action coupling:*** *Only A fails to factor: $A^{int} \neq A_\alpha \otimes A_\beta$*
2. ***Perception coupling:*** *P fails to factor: $P^{int} \neq P_\alpha \otimes P_\beta$*
3. ***Decision coupling:*** *D fails to factor: $D^{int} \neq D_\alpha \otimes D_\beta$*

Action coupling represents agents affecting each other's environments; perception and decision coupling represent direct internal access.

Definition 4.5 (Blanket-Mediated Interaction Condition)**.** *An interacting composition $\alpha \circledast \beta$ satisfies* BMIC *if:*

(BMIC-1) Perception factors: $P_{\alpha\circledast\beta} = P_\alpha \otimes P_\beta$

(BMIC-2) Decision factors: $D_{\alpha\circledast\beta} = D_\alpha \otimes D_\beta$

(BMIC-3) Action may couple: $A_{\alpha\circledast\beta}$ *arbitrary (subject to normalization)*

Remark. *BMIC captures the intuition that each agent's internal processing (perception $\to$ decision) is self-contained. Interaction occurs only through the external world: α affects β by changing the world; β perceives the changed world through its own sensory channel.*

Theorem 4.6 (BMIC Implies Compositional Correspondence)**.** *Let $\alpha \circledast \beta$ satisfy BMIC. Then:*

(i) The partition structure of $\Theta(\alpha \circledast \beta)$ factors: $\mu_{\alpha\circledast\beta} = \mu_\alpha \times \mu_\beta$, and similarly for sensory, active, and external states.

(ii) Each agent's Markov blanket property is preserved:

$$P(x'_\alpha \mid x_\alpha, s_\alpha, g_\alpha, w_\alpha, x_\beta, s_\beta, g_\beta, w_\beta) = P(x'_\alpha \mid x_\alpha, s_\alpha)$$

(iii) The combined system satisfies a joint blanket property:

$$P(x'_\alpha, x'_\beta \mid x_\alpha, x_\beta, B_\alpha, B_\beta, w_\alpha, w_\beta) = P(x'_\alpha, x'_\beta \mid x_\alpha, x_\beta, B_\alpha, B_\beta)$$

Proof. **(i)** Internal, active, and external states factor by construction. For sensory states: by BMIC-1, perception factors as $P_{\alpha\circledast\beta} = P_\alpha \otimes P_\beta$. By the same argument as Theorem 4.3, sensory equivalence factors:

$$(w_\alpha, w_\beta) \sim_{(x_\alpha, x_\beta)} (w'_\alpha, w'_\beta) \iff w_\alpha \sim_{x_\alpha} w'_\alpha \text{ and } w_\beta \sim_{x_\beta} w'_\beta$$

(ii) The dynamics kernel is:

$$K_{\alpha\circledast\beta} = P_\alpha[w_\alpha, x_\alpha, x'_\alpha] \cdot P_\beta[w_\beta, x_\beta, x'_\beta] \cdot D_\alpha[x'_\alpha, g'_\alpha] \cdot D_\beta[x'_\beta, g'_\beta] \cdot A^{\text{int}}[\ldots]$$

Marginalizing over $(x'_\beta, g'_\alpha, g'_\beta, w'_\alpha, w'_\beta)$:

$$P(x'_\alpha \mid \text{full state}) = P_\alpha[w_\alpha, x_\alpha, x'_\alpha] \cdot \underbrace{\int P_\beta \, dx'_\beta}_{=1} \cdot \underbrace{\int D_\alpha \, dg'_\alpha}_{=1} \cdot \underbrace{\int D_\beta \, dg'_\beta}_{=1} \cdot \underbrace{\int A^{\text{int}} \, d(w'_\alpha, w'_\beta)}_{=1}$$

Thus $P(x'_\alpha \mid \text{full state}) = P_\alpha[w_\alpha, x_\alpha, x'_\alpha]$, which depends on w_α only through $s_\alpha(w_\alpha, x_\alpha)$.

(iii) Since $P_{\alpha\circledast\beta} = P_\alpha \otimes P_\beta$ factors, $P(x'_\alpha, x'_\beta \mid \cdot) = P(x'_\alpha \mid x_\alpha, s_\alpha) \cdot P(x'_\beta \mid x_\beta, s_\beta)$. □

Theorem 4.7 (BMIC Failure Implies Blanket Fusion)**.** *If $\alpha \circledast \beta$ violates BMIC, then α and β cannot be represented as separate Markov blanket systems:*

(i) ***Perception coupling (BMIC-1 fails):*** *β's sensory equivalence depends on α's state, so s_β is not well-defined independently.*

(ii) ***Decision coupling (BMIC-2 fails):*** *β's action depends directly on α's experience, so B_β does not screen off β's dynamics.*

In either case, $\Theta(\alpha \circledast \beta)$ has a single merged blanket that cannot be factored.

Proof. **Case (i):** Suppose BMIC-1 fails. Then $P^{\text{int}}_\beta[w_\alpha, w_\beta, x_\beta, x'_\beta] \neq P^{\text{int}}_\beta[w'_\alpha, w_\beta, x_\beta, x'_\beta]$ for some $w_\alpha \neq w'_\alpha$. Define the w_α-conditional sensory equivalence:

$$w_\beta \sim^{w_\alpha}_{x_\beta} \tilde{w}_\beta \iff P^{\text{int}}_\beta[w_\alpha, w_\beta, x_\beta, \cdot] = P^{\text{int}}_\beta[w_\alpha, \tilde{w}_\beta, x_\beta, \cdot]$$

This equivalence relation depends on w_α. There is no single partition of W_β that serves as "β's sensory states" independently of α. The agents share a merged perceptual boundary.

Case (ii): Suppose BMIC-2 fails. Then $D_\beta^{\text{int}}[x_\alpha, x_\beta, g_\beta] \neq D_\beta[x_\beta, g_\beta]$—$\beta$'s decision depends on α's experience. The action g'_β depends on x'_α, which is α's internal state. The blanket B_β no longer fully characterizes β's interface: part of β's output is controlled by α's internal state, bypassing β's own decision process. □

Theorem 4.8 (Complete BMIC Characterization)**.** *Let α and β be conscious agents and $\alpha \circledast \beta$ an interacting composition. Then:*

1. ***(Sufficiency)*** *BMIC $\Rightarrow$ compositional correspondence holds (Theorem 4.6)*
2. ***(Necessity)*** *$\neg$BMIC $\Rightarrow$ agents fuse into a single MB system (Theorem 4.7)*
3. ***(Characterization)*** *BMIC is the exact boundary between dissociated (two separate MB systems) and associated (one unified MB system)*

Example 4.9 (BMIC Failure: Direct Internal Coupling)**.** *Let $X_\alpha = X_\beta = \{0, 1\}$ with perception coupling P_β^{int} depending on x_α:*

$$P_\beta^{int}[w_\alpha, w_\beta, x_\beta, x'_\beta] = \begin{cases} \delta_{w_\beta}(x'_\beta) & \text{if } x_\alpha = 0 \\ \delta_{1-w_\beta}(x'_\beta) & \text{if } x_\alpha = 1 \end{cases}$$

Agent α's internal state determines whether β perceives the world veridically or inverted. BMIC-1 fails: sensory equivalence for β depends on x_α. The agents have fused into a unified system.

Corollary 4.10 (Association Criterion)**.** *Agents* associate *(combine into unified consciousness) precisely when their interaction violates BMIC—when one agent's internal states directly influence another's internal processing, bypassing the sensory interface.*

5 Inverse Mapping

Not every Markov blanket system arises from a conscious agent. This section characterizes when the inverse mapping exists.

Theorem 5.1 (Representability)**.** *A Markov blanket system $M = (\Omega, \mu, s, a, \eta, K)$ admits representation as $\Theta(\alpha)$ for some conscious agent α if and only if:*

1. ***State space structure:*** *$\Omega \cong X \times G \times W$ for measurable spaces X, G, W*
2. ***Partition alignment:*** *$\mu \cong X$, $a \cong G$, $\eta \cong W$*
3. ***Kernel factorization:*** *K factors as $K = P \cdot D \cdot A$ where:*
   - *$P : W \times X \to \Delta(X)$ (perception-like)*
   - *$D : X \to \Delta(G)$ (decision-like)*
   - *$A : G \times W \to \Delta(W)$ (action-like)*

Proof. **(Necessity)** If $M = \Theta(\alpha)$ for $\alpha = (X, G, W, P, D, A)$, then by Definition 2.8: $\Omega_\alpha = X \times G \times W$, the partition aligns as specified, and $K_\alpha = P \cdot D \cdot A$ by equation (1).

(Sufficiency) Given conditions (1)–(3), define $\alpha = (X, G, W, P, D, A)$ where the kernels are from the factorization. This is a valid conscious agent by Definition 2.4, and $\Theta(\alpha) = M$ by construction. □

Definition 5.2 (Factorizability Obstruction)**.** *For MB system M, the* factorizability obstruction *is:*

$$Obs(M) = \inf_{P,D,A} \mathrm{D_{KL}}(K \| P \cdot D \cdot A)$$

where the infimum is over all valid kernel factorizations respecting the partition structure.

Corollary 5.3. *$Obs(M) = 0$ if and only if M is representable as $\Theta(\alpha)$ for some conscious agent α.*

Remark. *The kernel factorization $P \cdot D \cdot A$ encodes a specific causal structure: perception $\to$ decision $\to$ action. Systems with actions directly affecting experiences, or external states directly affecting actions without passing through perception, have $Obs(M) > 0$. Not all Markov blanket systems are "agential."*

5.1 Explicit Characterization via Conditional Independence

For finite state spaces, the factorizability condition admits an explicit characterization in terms of conditional independence constraints on the transition kernel.

Definition 5.4 (Factorizability Conditions)**.** *Let K be a transition kernel on $\Omega = X \times G \times W$ with $|X| = m$, $|G| = n$, $|W| = p$. Define the following conditions:*

(N1) Action independence of perception: *For all $x, x' \in X$, $w \in W$, and $g_1, g_2 \in G$:*

$$\sum_{g',w'} K[(x, g_1, w) \to (x', g', w')] = \sum_{g',w'} K[(x, g_2, w) \to (x', g', w')]$$

(N2) Markov decision: *For all $x' \in X$, $g' \in G$, the conditional*

$$\frac{\sum_{w'} K[(x, g, w) \to (x', g', w')]}{\sum_{g'',w'} K[(x, g, w) \to (x', g'', w')]}$$

is independent of (x, g, w) whenever the denominator is positive.

(N3) Action mediation (marginal): *For all $g' \in G$, $w, w' \in W$, the conditional*

$$\frac{\sum_{x'} K[(x, g, w) \to (x', g', w')]}{\sum_{x',w''} K[(x, g, w) \to (x', g', w'')]}$$

is independent of (x, g) whenever the denominator is positive.

(N4) Action mediation (conditional): *For all x', g', w, w':*

$$\frac{K[(x, g, w) \to (x', g', w')]}{\sum_{w''} K[(x, g, w) \to (x', g', w'')]} = \frac{\sum_{x''} K[(x, g, w) \to (x'', g', w')]}{\sum_{x'',w''} K[(x, g, w) \to (x'', g', w'')]}$$

whenever the left side is well-defined.

Theorem 5.5 (Characterization of Factorizability)**.** *Let K be a transition kernel on $\Omega = X \times G \times W$ with finite state spaces. Then K admits a factorization $K = P \cdot D \cdot A$ if and only if K satisfies conditions* (N1)–(N4)*.*

Proof. **(Necessity)** Suppose $K = P \cdot D \cdot A$. For (N1):

$$\sum_{g',w'} K[(x, g, w) \to (x', g', w')] = P[w, x, x'] \sum_{g'} D[x', g'] \sum_{w'} A[g', w, w'] = P[w, x, x']$$

which is independent of g. For (N2): $\sum_{w'} K[\cdot] / \sum_{g'',w'} K[\cdot] = P \cdot D/P = D[x', g']$, independent of (x, g, w). For (N3): the ratio reduces to $A[g', w, w']$, independent of (x, g). For (N4): $K/\sum_{w''} K = P \cdot D \cdot A/(P \cdot D) = A[g', w, w']$, matching the (N3) expression.

(Sufficiency) Assume (N1)–(N4). Define:

$$P[w, x, x'] := \sum_{g',w'} K[(x, g_0, w) \to (x', g', w')] \qquad \text{(independent of } g_0 \text{ by N1)}$$

$$D[x', g'] := \frac{\sum_{w'} K[(x_0, g_0, w_0) \to (x', g', w')]}{P[w_0, x_0, x']} \qquad \text{(independent of } (x_0, g_0, w_0) \text{ by N2)}$$

$$A[g', w, w'] := \frac{\sum_{x'} K[(x_0, g_0, w) \to (x', g', w')]}{\sum_{x',w''} K[(x_0, g_0, w) \to (x', g', w'')]} \qquad \text{(independent of } (x_0, g_0) \text{ by N3)}$$

These are valid Markov kernels (normalization follows from the definitions). From (N2), $\sum_{w'} K[\cdots \to (x', g', w')] = P[w, x, x'] \cdot D[x', g']$. By (N4):

$$K[(x, g, w) \to (x', g', w')] = \left(\sum_{w''} K[\cdots \to (x', g', w'')] \right) \cdot A[g', w, w'] = P[w, x, x'] \cdot D[x', g'] \cdot A[g', w, w']$$

□

Corollary 5.6 (Conditional Independence Reformulation)**.** *K is factorizable if and only if the following conditional independences hold in the joint distribution over (x, g, w, x', g', w'):*

1. $x' \perp g \mid (x, w)$
2. $g' \perp (x, g, w) \mid x'$
3. $w' \perp (x, g, x') \mid (g', w)$

Remark (Physical Interpretation)**.** *The four conditions encode a specific causal architecture: (N1) perception does not depend on current action—the senses are "passive"; (N2) decisions depend only on current experience, not on how that experience arose—no "hidden memory"; (N3–N4) the agent's effect on the world is fully mediated by its action—the world cannot "see inside" the agent. Each condition is independently necessary: examples exist satisfying any three while violating the fourth (see companion materials).*

6 Connection to Integrated Information

Definition 6.1 (Unit Agent)**.** *The* unit agent *is $\mathbf{1} = (\{*\}, \{*\}, \{*\}, P_*, D_*, A_*)$ where all state spaces are singletons and all kernels are the unique probability measures on singleton spaces. For any agent α, we have $\alpha \otimes \mathbf{1} \cong \alpha \cong \mathbf{1} \otimes \alpha$.*

Definition 6.2 (Agent Integration)**.** *For conscious agent α with finite state spaces, the* agent integration *is:*

$$\Phi_{\mathbf{Con}}(\alpha) = \inf_{(\alpha_1, \alpha_2)} \mathrm{D}_{\mathrm{KL}}(K_\alpha \| K_{\alpha_1} \otimes K_{\alpha_2})$$

where the infimum is over all non-trivial factorizations $\alpha \cong \alpha_1 \otimes \alpha_2$ (neither α_i is the unit agent $\mathbf{1}$ from Definition 6.1). For finite state spaces, the set of factorizations is finite (bounded by the number of partitions of each state space), so the infimum is achieved.

Theorem 6.3 (Φ-Dissociability Correspondence)**.** *For conscious agent α:*

$$\Phi_{\mathbf{Con}}(\alpha) = 0 \iff \alpha \text{ is dissociable} \iff \alpha \cong \alpha_1 \otimes \alpha_2 \text{ for non-trivial } \alpha_1, \alpha_2$$

Proof. ($\Rightarrow$) If $\Phi_{\mathbf{Con}}(\alpha) = 0$, then for finite state spaces the infimum is achieved at some (α_1, α_2) with $\mathrm{D}_{\mathrm{KL}}(K_\alpha \| K_{\alpha_1} \otimes K_{\alpha_2}) = 0$. Since $\mathrm{D}_{\mathrm{KL}}(P \| Q) = 0$ iff $P = Q$ a.e., we have $K_\alpha = K_{\alpha_1} \otimes K_{\alpha_2}$, establishing $\alpha \cong \alpha_1 \otimes \alpha_2$.

($\Leftarrow$) If $\alpha \cong \alpha_1 \otimes \alpha_2$, then $K_\alpha = K_{\alpha_1} \otimes K_{\alpha_2}$ (up to canonical isomorphism), so $\mathrm{D}_{\mathrm{KL}}(K_\alpha \| K_{\alpha_1} \otimes K_{\alpha_2}) = 0$. □

Corollary 6.4. *BMIC failure for interacting agents α, β indicates $\Phi_{\mathbf{Con}}(\alpha \circledast \beta) > 0$: the combined system is integrated.*

Remark (Connection to IIT)**.** *The integration measure $\Phi_{\mathbf{Con}}$ is related to, but distinct from, Tononi's Φ_{IIT} [9]. The key differences: (i) $\Phi_{\mathbf{Con}}$ optimizes over agent-structure-respecting factorizations (both factors must have $P \cdot D \cdot A$ structure), while Φ_{IIT} optimizes over all bipartitions of the state space; (ii) the optimization domains differ, so the measures are not directly comparable as inequalities; (iii) we use KL divergence throughout, whereas IIT 4.0 employs earth mover's distance. Despite these differences, qualitative equivalence holds for any proper divergence: $\Phi = 0$ if and only if the system is dissociable. The compositional correspondence (Theorem 4.3) holds precisely when $\Phi_{\mathbf{Con}} = 0$; it fails precisely when $\Phi_{\mathbf{Con}} > 0$. Integration—non-factorizability of the dynamics—is the obstruction to compositional correspondence.*

6.1 Dynamical versus Agent Dissociability

The definition of $\Phi_{\mathbf{Con}}$ requires factors to be conscious agents. We now clarify the relationship between two notions of factorizability.

Definition 6.5 (Dynamical Dissociability)**.** *Agent α is* dynamically dissociable *if there exist kernels K_1, K_2 on non-trivial state spaces Ω_1, Ω_2 with $\Omega_\alpha \cong \Omega_1 \times \Omega_2$ such that:*

$$K_\alpha = K_1 \otimes K_2$$

Definition 6.6 (Aligned Factorization)**.** *A factorization $\Omega_\alpha = \Omega_1 \times \Omega_2$ is* aligned *with agent structure if:*

1. $\Omega_1 = X_1 \times G_1 \times W_1$ *and* $\Omega_2 = X_2 \times G_2 \times W_2$
2. $X = X_1 \times X_2$, $G = G_1 \times G_2$, $W = W_1 \times W_2$

Theorem 6.7 (Structural Marginal Characterization)**.** *Let $\alpha = (X, G, W, P, D, A)$ be a conscious agent. The following are equivalent:*

(i) α is agent-dissociable: $\alpha \cong \alpha_1 \otimes \alpha_2$

(ii) α is dynamically dissociable via an aligned factorization, and each kernel factors: $P = P_1 \otimes P_2$, $D = D_1 \otimes D_2$, $A = A_1 \otimes A_2$

Proof. (i) $\Rightarrow$ (ii): If $\alpha = \alpha_1 \otimes \alpha_2$ with $\alpha_i = (X_i, G_i, W_i, P_i, D_i, A_i)$, then by definition of the tensor product: $X = X_1 \times X_2$, $G = G_1 \times G_2$, $W = W_1 \times W_2$, and $P = P_1 \otimes P_2$, $D = D_1 \otimes D_2$, $A = A_1 \otimes A_2$. Thus $K_\alpha = K_{\alpha_1} \otimes K_{\alpha_2}$ with aligned factorization.

(ii) $\Rightarrow$ (i): Given factored kernels P_i, D_i, A_i, define $\alpha_i = (X_i, G_i, W_i, P_i, D_i, A_i)$. Each α_i is a well-defined conscious agent, and $\alpha = \alpha_1 \otimes \alpha_2$ by construction. □

Corollary 6.8 (Non-Aligned Factorizations)**.** *If α is dynamically dissociable via a non-aligned factorization, the factors lack agent structure.*

Example 6.9 (Dynamically but Not Agent-Dissociable)**.** *Let $X = G = W = \{0, 1\}$ with uniform kernels: $P[w, x, x'] = D[x, g] = A[g, w, w'] = \frac{1}{2}$. Then $K_\alpha[(x, g, w) \to (x', g', w')] = \frac{1}{8}$ uniformly.*

This kernel factors as $K_\alpha = K_{XG} \otimes K_W$ where K_{XG} is uniform on $X \times G$ and K_W is uniform on W. However, this factorization groups experience and action together: the factor K_{XG} has no world space and cannot be a conscious agent. The factorization is non-aligned.

The only aligned factorization would require X, G, W to each factor, but $|\{0, 1\}| = 2$ admits no non-trivial factorization. Thus α is dynamically dissociable but not agent-dissociable.

Remark. *This distinction has philosophical import: dynamical dissociability means behavior factors; agent-dissociability means experiential perspectives factor. A system can exhibit independent behavioral subsystems without those subsystems constituting separate experiencers.*

7 Non-Genericity Analysis

The conditional independence in Theorem 3.1 is *non-generic*: it does not hold for arbitrary dynamical systems. This section provides precise characterizations.

Proposition 7.1 (Codimension)**.** *Let $\Omega = \mu \cup B \cup \eta$ with $|\mu| = m$, $|B| = b$, $|\eta| = e$, and $n = m + b + e$. The set of $n \times n$ stochastic matrices satisfying the Markov blanket property has codimension at least $m^2 \cdot b \cdot (e - 1)$ in the space of all stochastic matrices.*

Proof. The Markov blanket property requires that for all $(\mu, b) \in S_\mu \times S_B$ and all $\mu' \in S_\mu$:

$$\sum_{b', \eta'} K[(\mu, b, \eta_1) \to (\mu', b', \eta')] = \sum_{b', \eta'} K[(\mu, b, \eta_2) \to (\mu', b', \eta')] \quad \forall \eta_1, \eta_2 \in S_\eta$$

This imposes $(e-1)$ independent linear constraints for each (μ, b, μ') triple. The total number of constraints is $m \cdot b \cdot m \cdot (e-1) = m^2 \cdot b \cdot (e-1)$. □

Proposition 7.2 (Graph-Theoretic Characterization)**.** *A network dynamical system on graph $G = (V, E)$ supports a Markov blanket partition (μ, B, η) if and only if B is a (μ, η)-vertex separator: every path from μ to η passes through B.*

Proof. ($\Rightarrow$) If B is not a separator, there exists an edge (v, u) with $v \in \mu$ and $u \in \eta$. Then η can directly affect μ' through this edge, violating conditional independence.

($\Leftarrow$) If B separates μ from η, any influence of η on μ' must pass through B. When dynamics respect graph structure (only neighbors interact), the MB property holds. □

Proposition 7.3 (Failure in Standard Systems)**.** *The following system classes do not generically satisfy the Markov blanket property:*

System Class	*MB Property*	*Reason*
Conscious Agents	*Yes*	*$P \cdot D \cdot A$ factorization*
MB Systems	*Yes*	*By definition*
Cellular Automata	*No*	*Blanket affected by exterior*
Generic Markov Chains	*No (measure 0)*	*Random matrices don't factor*
Reaction-Diffusion	*No*	*Diffusive coupling*
Network Dynamics	*No (generically)*	*No screening topology*

Proof sketch (Cellular Automata). In a 1D cellular automaton with cell c internal, cells $\{c-1, c+1\}$ as blanket, and cells $\{c-2, c+2, \ldots\}$ external: the blanket update $\sigma'(c-1) = f(\sigma(c-2), \sigma(c-1), \sigma(c))$ depends directly on external cell $c-2$. The blanket itself is affected by external states, so $P(\mu' \mid B, \eta) \neq P(\mu' \mid B)$ in general. □

Corollary 7.4 (Biological Relevance)**.** *Neural networks exhibit approximate Markov blanket structure due to hierarchical organization and modular connectivity. This is non-generic: in Erdős-Rényi random graphs $G(n, p)$ with $p > \log(n)/n$, no non-trivial vertex separator exists with high probability.*

Proposition 7.5 (Non-Genericity of Agent Structure)**.** *Let $|X| = m$, $|G| = n$, $|W| = p$. Consider the space of "generalized agents" with an unfactored kernel $K : (W \times X \times G) \to \Delta(X \times G \times W)$. The $P \cdot D \cdot A$ factorized agents form a submanifold of strictly lower dimension:*

- *Full kernel:* $\dim = mnp \cdot (mnp - 1)$ *parameters*
- *$P \cdot D \cdot A$ factorized:* $\dim = mp(m-1) + m(n-1) + np(p-1)$ *parameters*

For the minimal case $m = n = p = 2$: full dimension $= 56$, factorized dimension $= 10$, codimension $= 46$.

Proof. Count free parameters. A general kernel from mnp states to $\Delta(mnp)$ has $mnp \cdot (mnp-1)$ free parameters (each row of the stochastic matrix is a probability vector with $mnp - 1$ degrees of freedom).

For the factorized kernel $K = P \cdot D \cdot A$:

- $P \in \mathcal{K}(W \times X, X)$: each of mp source states maps to $\Delta(m)$, giving $mp(m-1)$
- $D \in \mathcal{K}(X, G)$: each of m source states maps to $\Delta(n)$, giving $m(n-1)$
- $A \in \mathcal{K}(G \times W, W)$: each of np source states maps to $\Delta(p)$, giving $np(p-1)$

Total: $mp(m-1) + m(n-1) + np(p-1)$. Since $mp(m-1) + m(n-1) + np(p-1) < mnp(mnp-1)$ for all $m, n, p \geq 2$, the factorized agents have positive codimension. By Sard's theorem, they form a measure-zero set. □

Remark. *This strengthens the non-genericity analysis: not only is the Markov blanket property rare (Proposition 7.1), but the $P \cdot D \cdot A$ factorization that guarantees it is itself a severe constraint. Agent-like dynamics occupy a thin submanifold of all possible dynamics on the same state space.*

8 Discussion

8.1 What the Correspondence Establishes

We have shown that conscious agent structure is *sufficient* for Markov blanket structure: the P-D-A factorization implies conditional independence. This is a derived result—Hoffman's framework was motivated by perception-action cycles, not statistical independence. That it implies the Markov property reveals deep structural compatibility.

The correspondence is *not* an equivalence. Not every Markov blanket system admits conscious agent structure (Theorem 5.1). The factorization $K = P \cdot D \cdot A$ imposes a specific causal ordering that not all MB systems possess.

8.2 Significance of Non-Genericity

The codimension bounds (Proposition 7.1) establish that conditional independence is a genuine constraint, not a trivial property. For a system with $m = b = e = 10$, the codimension is at least $100 \cdot 10 \cdot 9 = 9000$. The MB-satisfying matrices form a measure-zero set.

This result should be read alongside Fields' [2] argument that Markov blankets are *generic* physical interaction surfaces: any interaction between spatially separated systems occurs across a boundary satisfying the conditional independence property. These results are complementary, not contradictory. Fields' genericity holds among physical systems, where locality of interactions guarantees the screening-off property. Our non-genericity holds in the space of all mathematically possible dynamical systems, which need not respect locality. The conjunction is informative: physics selects for Markov blanket structure despite its mathematical non-genericity, suggesting that physical laws impose strong organizational constraints.

This has three implications:

1. The correspondence captures real structure, not an artifact
2. Systems exhibiting Markov blankets (biological or artificial) have non-generic organization
3. The physical genericity of Markov blankets [2] despite their mathematical non-genericity points to something special about physical law

8.3 Limitations

1. **Correspondence $\neq$ equivalence:** We show compatibility, not identity of frameworks
2. **Non-genericity $\neq$ significance:** Measure-zero establishes specificity, not importance
3. **Discrete-time:** Extension to continuous-time stochastic processes requires care
4. **Finite-dimensional:** The codimension bounds assume finite state spaces

9 Conclusion

We have established a formal correspondence between Hoffman's Conscious Agents and Friston's Markov Blanket systems. The main theorem (Theorem 3.1) proves that the P-D-A kernel factorization implies conditional independence. The compositional theorems characterize when multi-agent systems preserve this correspondence: Theorem 4.3 handles non-interacting agents, while the BMIC theorems (Theorems 4.6–4.8) provide a complete characterization for interacting agents—BMIC is necessary and sufficient for maintained dissociation. The inverse mapping theorem (Theorem 5.1) characterizes when MB systems admit conscious agent structure. The Φ-dissociability theorem (Theorem 6.3) connects integration to Tononi's framework. The non-genericity analysis shows these are genuine structural constraints.

The correspondence reveals that two influential frameworks for understanding minds—one grounded in perception theory, one in statistical physics—share deep formal structure. Whether

this compatibility reflects something fundamental about consciousness or merely the constraints of representing agent-environment boundaries mathematically remains an open question.

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References

[1] Fields, C., Hoffman, D. D., Prakash, C., & Singh, M. (2018). Conscious agent networks: Formal analysis and application to cognition. *Entropy*, 20(3), 175.

[2] Fields, C. (2019). Markov blankets are general physical interaction surfaces. *BioSystems*, 182, 31–38.

[3] Fields, C., Friston, K., Glazebrook, J. F., & Levin, M. (2022). A free energy principle for generic quantum systems. *Progress in Biophysics and Molecular Biology*, 173, 36–59.

[4] Friston, K. (2019). A free energy principle for a particular physics. *arXiv:1906.10184*.

[5] Hoffman, D. D. (2019). *The Case Against Reality*. W. W. Norton.

[6] Hoffman, D. D., & Prakash, C. (2014). Objects of consciousness. *Frontiers in Psychology*, 5, 577.

[7] Kallenberg, O. (2021). *Foundations of Modern Probability* (3rd ed.). Springer.

[8] Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann.

[9] Tononi, G. (2015). Integrated information theory. *Scholarpedia*, 10(1), 4164.

[10] Parr, T., Pezzulo, G., & Friston, K. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.